

General Microbiology

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Course code: Mbio 1053

Objectives (summary)

Upon completion of this unit of instruction, the student will be able to:

- Discuss Medical Microbiology
- Discuss the historical background of Medical Microbiology
- Discuss the basic differences between Prokaryotic and eukaryotic cells
- Discuss the methods used for classification of Microorganisms
- Identify external and internal structural components of bacteria
- Discuss bacterial metabolism and growth

Introduction to Medical Microbiology

 **Microbiology**, study of microorganisms that are individually too small to be seen with the naked eye

- single cells (unicellular)
- cell clusters (multicellular)

☐ Study nature of their;

- Reproduction,
- Physiology,
- Participation in the process of nature,
- Relationship with other living organisms

 **Medical Microbiology**

 diagnosis, treatment, and prevention of infectious diseases caused by **all groups of microorganisms.**

Cont...

- Microorganisms have a tremendous impact on all life and the physical and chemical makeup of our planet.
- They are responsible for cycling the chemical elements essential for life, including;
 - Carbon, Nitrogen, Sulfur, Hydrogen, and Oxygen;
 - Photosynthesis is carried out by microorganisms than by green plants.
 - They also are a source of nutrients at the base of all ecological food chains and webs.

Cont...

- Humans also have an intimate relationship with Microorganisms
 - > 90% of the cells in our bodies are microbes
 - bacteria present in the average human gut weigh about **1 kg**
 - a human adult will **excrete** his or her weight in fecal bacteria each year.
 - 8% of the DNA is derived from remnants of **viral genomes**

Subdivision of microbiology

- ✓ **Bacteriology** study about bacteria.
- ✓ **Mycology** study about fungi.
- ✓ **Virology** study about viruses.
- ✓ **Immunology** study systems of body defenses that protect against infection.
- ✓ **Parasitology** study about parasites

History of Microbiology

- Mankind has always been affected by diseases that were originally believed to be visitations by **God** and meant to punish evildoers.
- At **117-26 BC** **Marcus Terentius Varro**; **disease** was caused by **animated** particles invisible to the naked eye **suspended in the air**
- In 1025 AD **Abū Alī ibn Sīnā** hypothesized that **tuberculosis** and other diseases might be contagious.
- **Fracastorius** (1500 G.C.) proposed that the agents of communicable disease were **living germs**
- All these early claims about the existence of microorganisms were **speculative** and were not based on any data or science.

Microscope and Discovery of Microorganisms

- **Antony Van Leeuwenhoek** (1632-1723 G.C.)
 - He was first to discover **Microorganisms** by using a simple microscope with one lens of his design (**invention**).
 - Observed protozoa from the gut of animal
 - Bacteria from teeth scraping
 - He was the first who properly described the different shapes of bacteria.
- After 100 years **Otto Müller** extended van Leeuwenhoek's studies and organized bacteria into **genera** and **species**
- 'where did the microbes were originated???' was main topic of discussion among scientist at that time

Cont...

- To get the answer and an explanation for spontaneous appearance of living things
 - from decaying meat
 - stagnating ponds,
 - fermenting grains,
 - infected wounds.
- two major theories were formulated on origins of microbe
 - **Theory of Abiogenesis**
 - **Theory of Biogenesis**

The theory of Abiogenesis

- Is also called **spontaneous generation**.
- **Aristotle** (384-322 BC): founder
- States that living things originated from non-living things.
- **Francesco Redi** (1626-1697) tried to set an experiment to **disprove** spontaneous generation.
- **Preformed an experiment**
 - He put the meat in a bottle and covered it with a gauze.
 - He observed that the flies laid eggs from which the maggots developed.
 - He said maggots did not develop from meat but from flies egg.

John Needham (1749)

- Performed experiments similar to Redi's on the origin of life in microscopic organisms
- Introduced the **first culture(broth) medium** for microbial growth.
- Broth put in flasks, some were sealed with corks, and some were not.
- All flasks became cloudy, result different from Redi's experiment.
- He suggested that life originate spontaneously from nonliving matters
- The spontaneous generation opponents didn't accept his conclusion,
 - they said it could be due to entrance organisms from air or flasks, improper seal.

Lazzaro Spallanzani (1776)

- Repeated Needham's experiments to disprove spontaneous generation in microscopic life.
- Boiled broth after placing in flasks.
- Sealed flasks by plugging with solid stopper.
- Results more consistent with Redi's.
- Occasionally sealed flask → cloudy.
- Not accepted by spontaneous generation supporters, because they said that heating may have destroyed, degraded "vital force" and air was not allowed to enter.

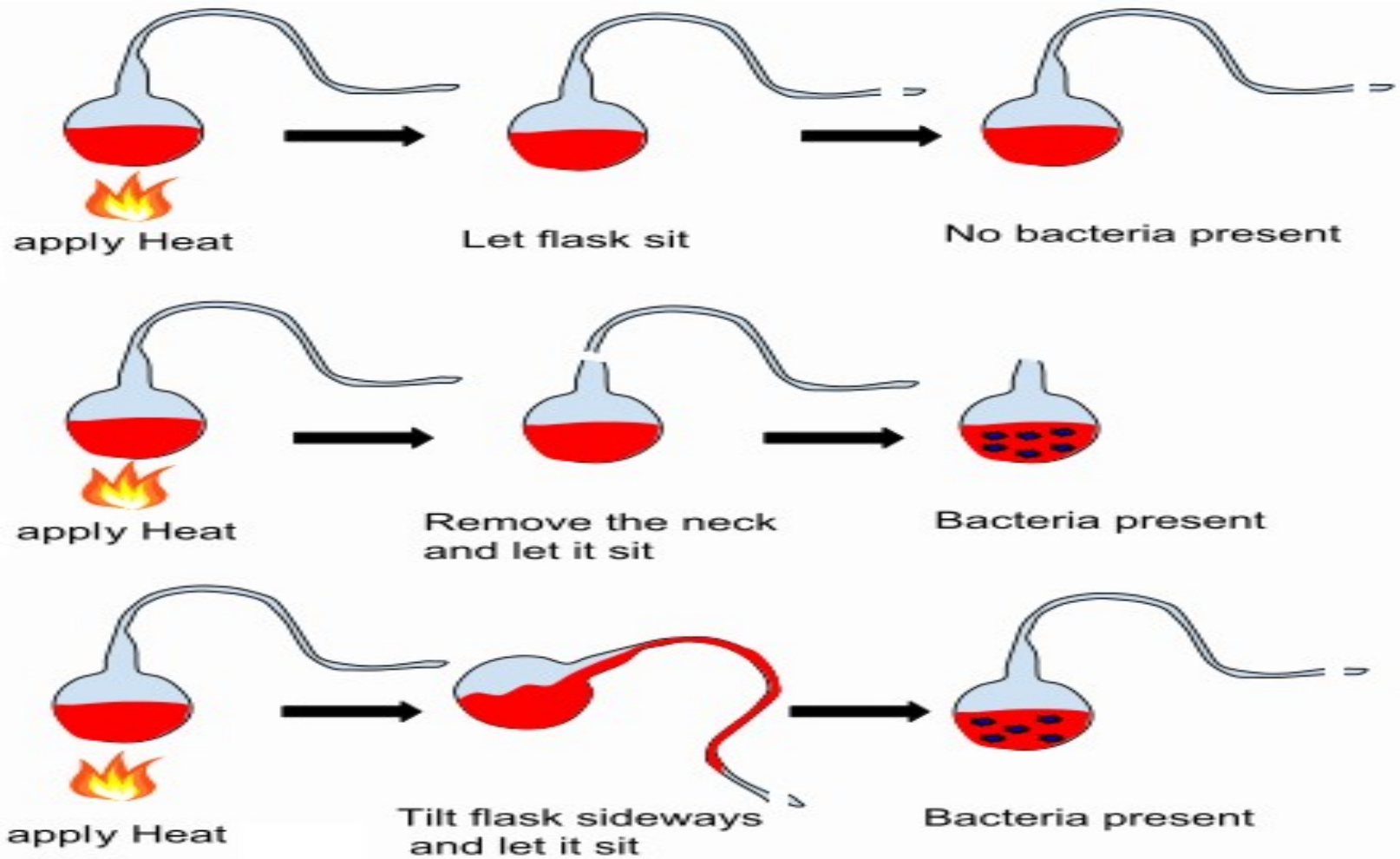
The theory of Biogenesis

- The theory states that life comes from pre-existing life or **biotic generation**.
- **Louis Pasteur** (1822-1895 GC) expanded Spallanzani's findings
- Performed an experiment to disprove the theory of spontaneous generation
- was the scientist who disproved the theory of abiogenesis once and for all.

Louis Pasteur experiment

- Broth put into flask with "S" shaped neck
- broth is boiled to kill all microorganisms.
- Microorganisms trying to enter the flask are stuck in the "S" curves of the neck
- Nothing grew in the broths in the course of Pasteur's experiment.
- This meant that the living organisms that grew in such broths came from outside, as spores on dust, rather than spontaneously generated
- Pasteur dealt the death blow to the theory of spontaneous generation

Pasteur Swan-Neck flask experiment (1861)



Major contribution of Louis Pasteur

- Microbial theory of fermentation
- Principles and practice of sterilization and pasteurization
- Control of diseases of silk worms
- Development of vaccines against anthrax and rabies.
- Discovery of streptococci

The Germ Theory of diseases

- Pasture has also developed the germ theory of diseases,
 - a specific disease is caused by a specific type of microorganism.
- **Robert Koch**, in 1876 established an experimental procedure to prove the germ theory of disease
 - The scientific procedure is known as **Koch's Postulate**.

Koch's Postulate: - proof of germ theory of disease

A Micro-organism can be accepted as a causative agent of an infectious disease only if the following conditions are satisfied.

1. The microorganism must be present in every case of the disease but absent from healthy host
2. The suspected microorganism must be isolated and grown in a pure culture from lesions of the disease.
3. The isolated organism, in pure culture, when inoculated in suitable laboratory animals should produce a similar disease.
4. It should be possible to re-isolate the bacterium in pure culture from the lesion produced in the experimental animal.

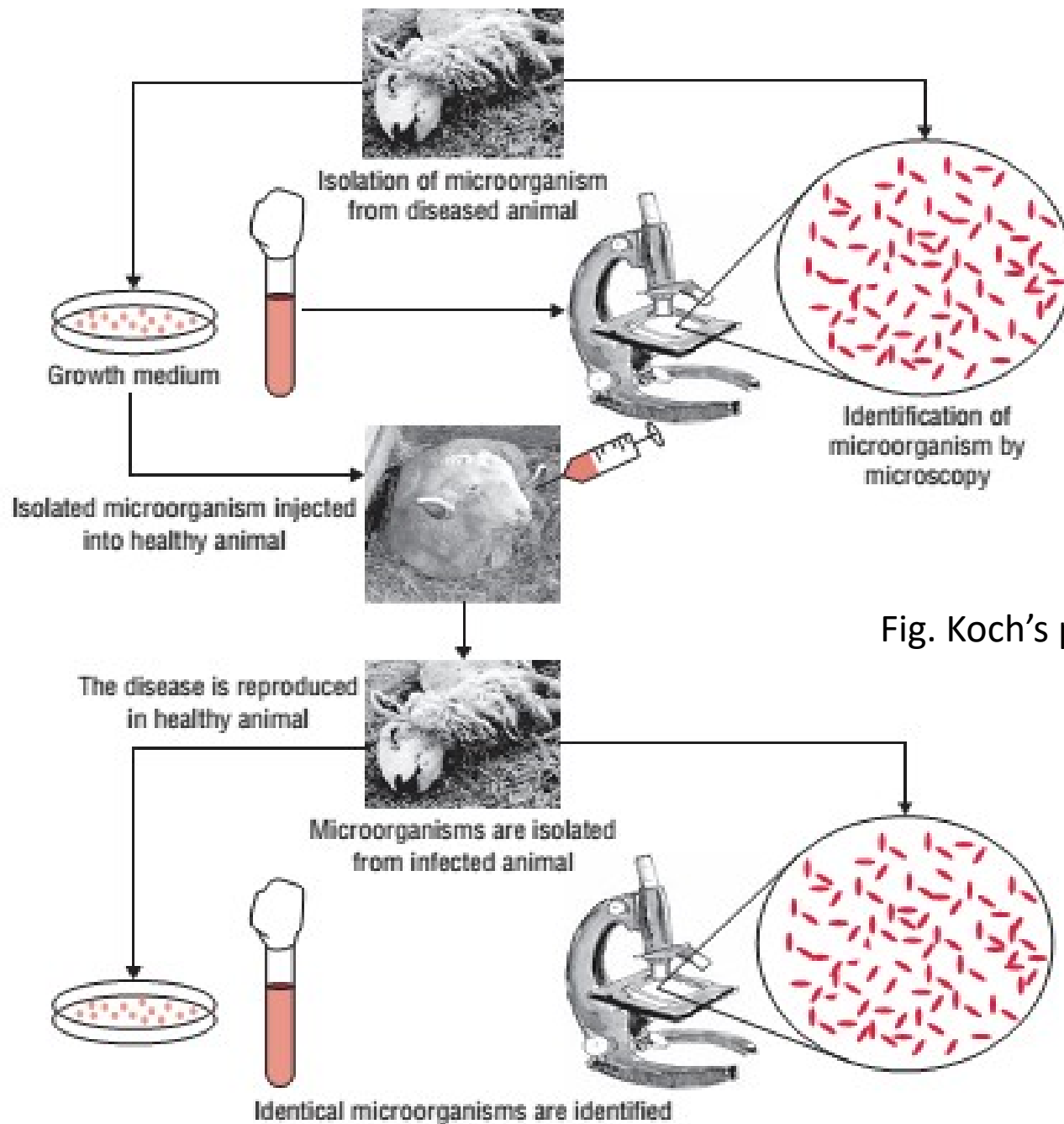


Fig. Koch's postulate

Exceptions to Koch's postulate

- Many healthy people carry pathogens but do not exhibit symptoms of the disease.
- Some microbes are very difficult or impossible to grow in vitro (in the laboratory) in artificial media. Eg. *Treponema pallidum*.
- Many pathogens are species specific. Eg. *Brucella abortus* cause abortion in animals but not in humans.
- Certain diseases develop only when an opportunistic pathogen invades immuno-compromised host.

Major achievements of Robert Koch

- Verification of germ theory of diseases
- Introduction of staining techniques for visualization of microorganisms
- Discovery and use of solid medium in bacteriology
- Discovery of causative agents of:
 - Tuberculosis
 - Cholera
 - Anthrax
- Koch's postulates
- Use of laboratory animals for experimental infections

Host-Parasite Relationship

- Human host is normally in contact with many microorganisms
(normal flora)
 - The normal flora provides some benefits to the host they live in
- The term host & parasite relationship is also known as SYMBIOSIS
means the coexistence of two organisms
- Symbiosis is two types – 1) Beneficial 2) harmful

Types of Symbiosis

- **Commensalism**- parasite deriving nourishment to the host without causing any harm or benefit to the host.
 - obtain nutrients, shelter, support, or locomotion from the host species
- **Parasitism**- the relationship may be in which one organism damage the other.
 - **Ectoparasites** live on the outside of the host's body, such as lice and ticks
 - **Endoparasites** present inside the host's body, such as nematodes and hookworms
- **Mutualism**- an association b/n two different species and both members of the association benefit
 - **Obligate** the relation matters their survival
 - **Facultative** both organisms benefit from but are not dependent upon their relationship for survival

Some important terms

- **Host:** human (or animal or others) that support the growth and survival and protection of the parasite
- **Parasite:** bacteria, viruses, fungi or parasites which live in or within the host ,may cause disease or live mutually with the host
- **Pathogen** a microorganism able to produce disease.
- **Pathogenicity** is the ability of a microorganism to cause disease in another organism
- **Resservoir** any person, plant, soil and substance in which an infectious agent normally lives and multiplies
- **virulance** degree of pathogenicity of microorganism to cause disease.
involves adherence, invasion and toxigenicity
- **Toxigenecity** the ability of microorganism to produce toxins.

Introduction to Bacteriology

Bacteria are single-celled microorganisms

- Lack a nuclear membrane and true nucleus
- Metabolically active but species of **Chlamydia** and **Rickettsia** are obligate intracellular organisms.
- Bacterial cells are between 0.1 -10 μm size.
- **cell wall (peptidoglycan)**
- devoid of mitochondria, endoplasmic reticulum (ER), or Golgi bodies

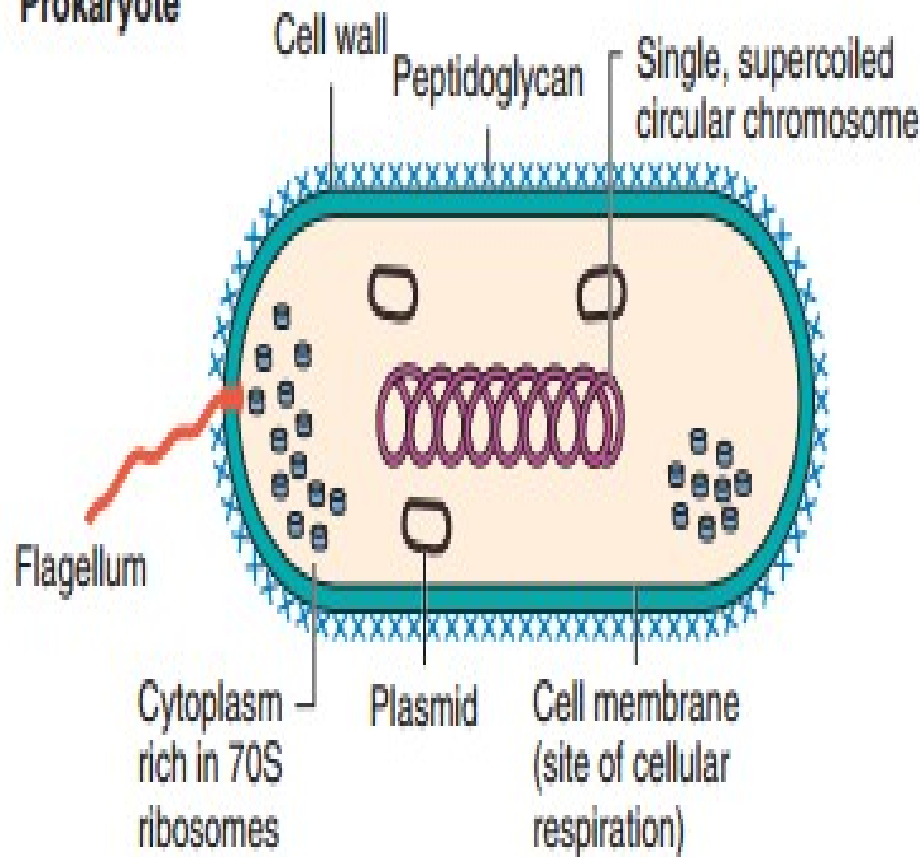
 The absence of the preceding bacterial cell structures differentiates them from **eukaryotes**.

- Superficially, bacteria appear to be relatively **simple** forms of life
 - In fact, they are **sophisticated** and **highly adaptable**.
 - Multiply at rapid rates, can utilize an enormous variety of **hydrocarbon** substrates, including **phenol**, **rubber**, and **petroleum**
 - Bacteria are highly successful colonizers,
 - Exist widely in both parasitic and free living forms

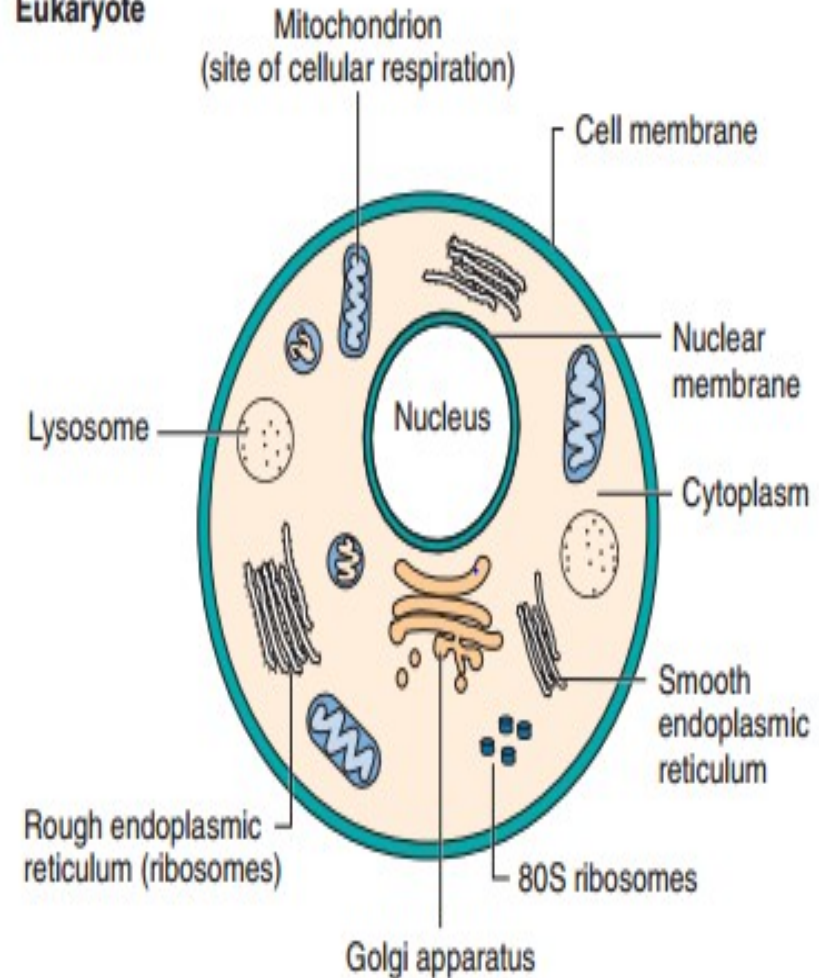
CHARACTERISTIC		PROKARYOTIC CELLS	EUKARYOTIC CELLS
Chromosome		Usually single, circular	Multiple
Nucleus		No nuclear envelope or nucleoli	Membrane bound, nucleoli present
Membrane-bound Organelles		Not present	Present
Cell wall		Usually present, many contain peptidoglycan	Present in plant cells, no peptidoglycan
Plasma membrane		No carbohydrates, most lack sterols	Sterols and carbohydrates present
Ribosome		70S	80S (70S in organelles)
Average size		0.2–2 mm in diameter	10–100 mm in diameter
Genome	location	In the nucleoid, at the mesosome	In the nucleus
	Chromosomal DNA	Circular; complexed with RNA	Linear; complexed with basic histones and other proteins
	Plasmid	Present	Absent
Reproduction		Asexual (binary fission)	Sexual and asexual

Major features of prokaryotes and eukaryotes

Prokaryote



Eukaryote



- Bacteria are **unicellular** microorganisms and generally not organized into tissues.
- Each bacterium grows and divides independently of any other bacterium
- Many bacteria can form aggregated structures called biofilms.
- Bacteria that have aggregated into biofilms can communicate information about population size and metabolic state
- This type of communication is called quorum sensing and operates by the production of small molecules called autoinducers or pheromones

General property

- Typical prokaryotic cell
- Contain both DNA and RNA
- Most grow in artificial media
- Replicate by binary fission
- Almost all contain rigid cell wall, with the exception of the **mycoplasma**
- Sensitive to antimicrobial agent
- All bacteria, pathogenic and saprophytic are unicellular

Morphological features of bacteria

- Bacteria are so small that their presence was only first recognized in 1677 with the aid of primitive microscope
- Now bacteria are usually examined under light microscopes 1,000X magnification
- Details of their internal structure can be observed more powerful transmission electron microscopes
- Bacteria have to be stained with a colored dye so that they will stand out from their background
- Bacteria can be seen clearly with the light microscope when appropriate methods are used

Cont....

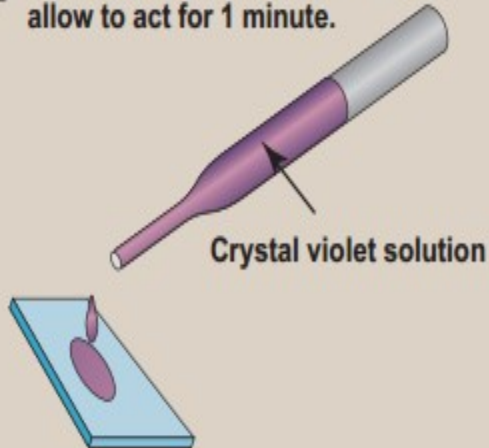
- 📖 Description of bacterial structural detail is challenging even with the best of light microscopes due to their
 - ✓ small size
 - ✓ are colorless and transparent and
 - ✓ have a refractive index similar to that of the surrounding liquid
- ✓ Common staining methods for bacteria examination
 - Wet-mount preparations
 - Gram stain
 - Acid-fast stains
 - Special stains

Gram stain

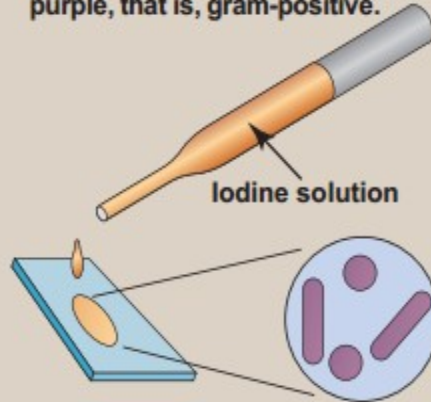
- Developed in 1884 by the Danish physician **Hans Christian Gram**
- differential staining method
- Gram-staining use **cell wall** composition for differentiation
- most commonly used, It places bacteria into one of the two main groups
 - Gram-positive; stain **purple/blue**
 - Gram-negative; stain **pink/red**
- The difference between the two is **permeability** of the cell wall to acetone and alcohol solvents
- Is a rapid, powerful, easy help to develop an initial diagnosis
- The Gram stain consists of gentle heat fixing (methyl alcohol)
- four sequential dyes are added
 - ✓ **crystal violet**
 - ✓ **Iodine**

Procedure of gram staining

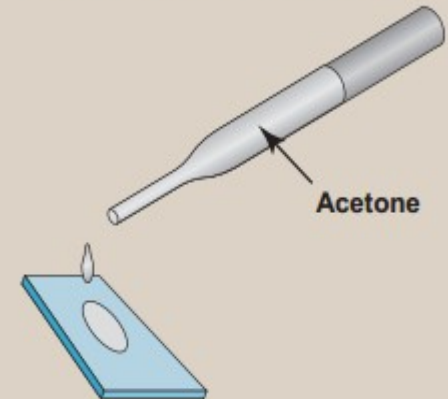
- 1** Heat-fix specimen to slide. Flood slide with crystal violet solution; allow to act for 1 minute.



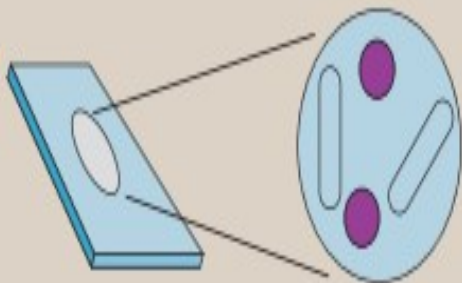
- 2** Rinse the slide, then flood with iodine solution; allow iodine to act for 1 minute. Before acetone decolorization (next step), all organisms appear purple, that is, gram-positive.



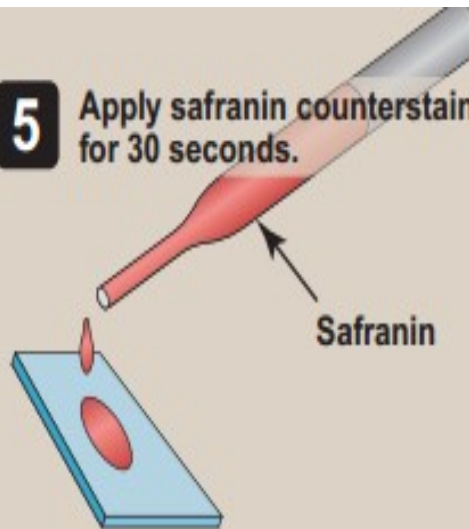
- 3** Rinse off excess iodine. Decolorize with acetone for approximately 5 seconds (time depends on density of specimen).



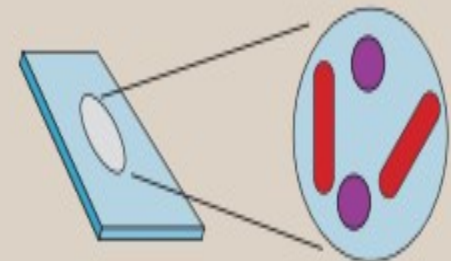
- 4** Wash slide immediately in water. After acetone decolorization, those organisms that are gram-negative are no longer visible.

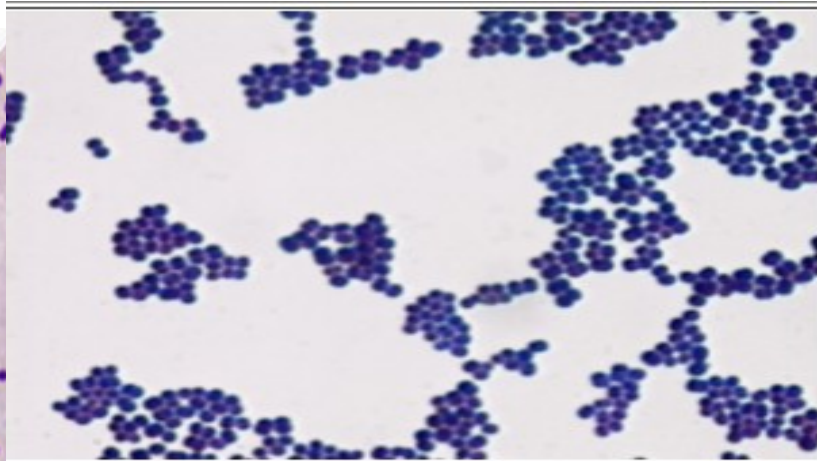


- 5** Apply safranin counterstain for 30 seconds.

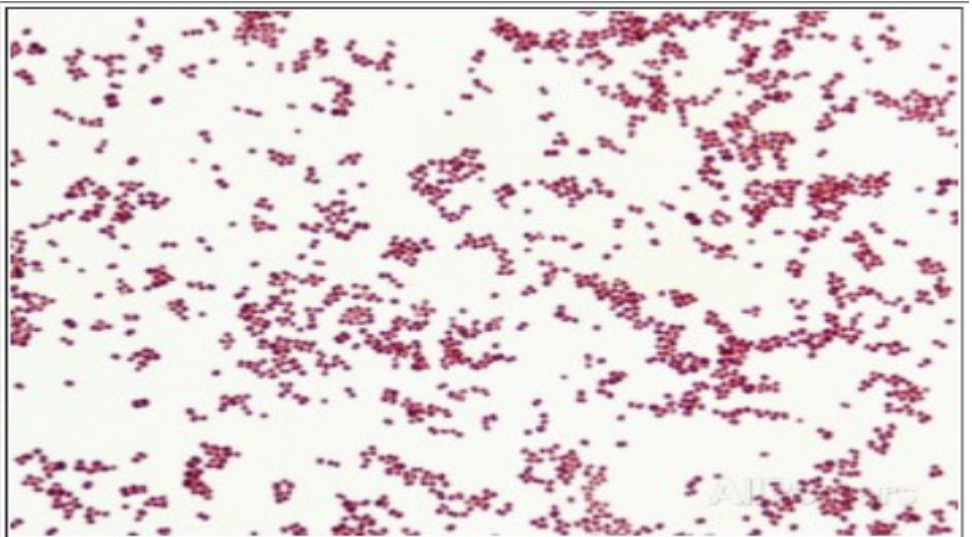


- 6** Wash in water, blot, and dry in air. Gram-negative organisms are visualized after application of the counterstain.





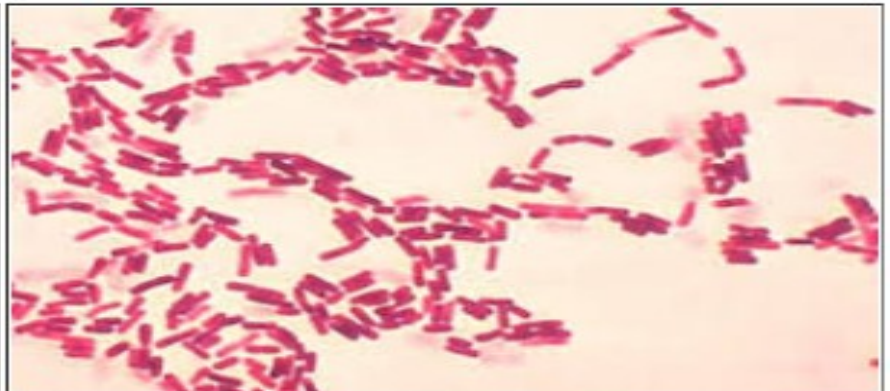
Gram positive cocci



Gram negative cocci



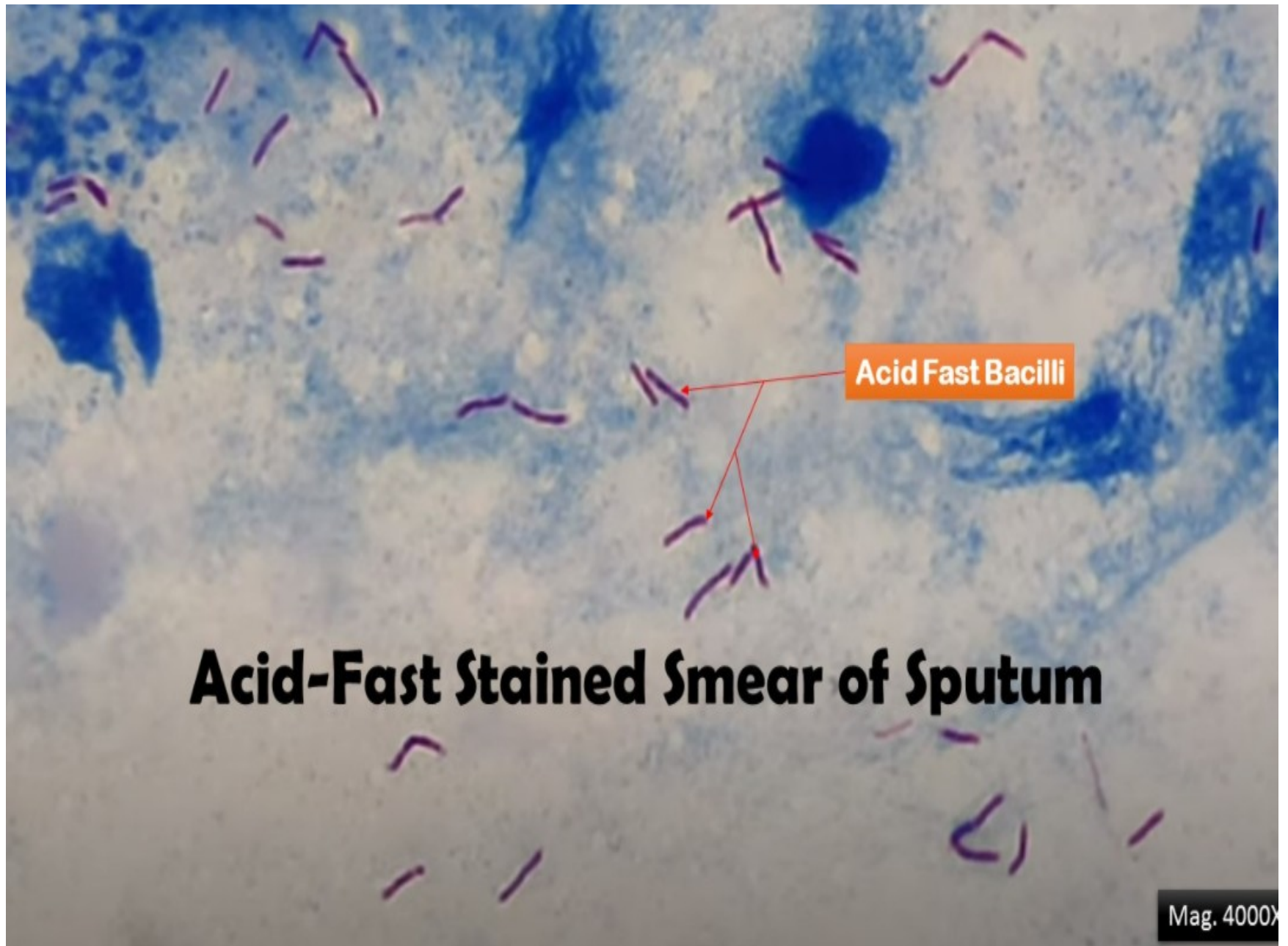
Gram positive rod



Gram negative rod

Acid-fast stains

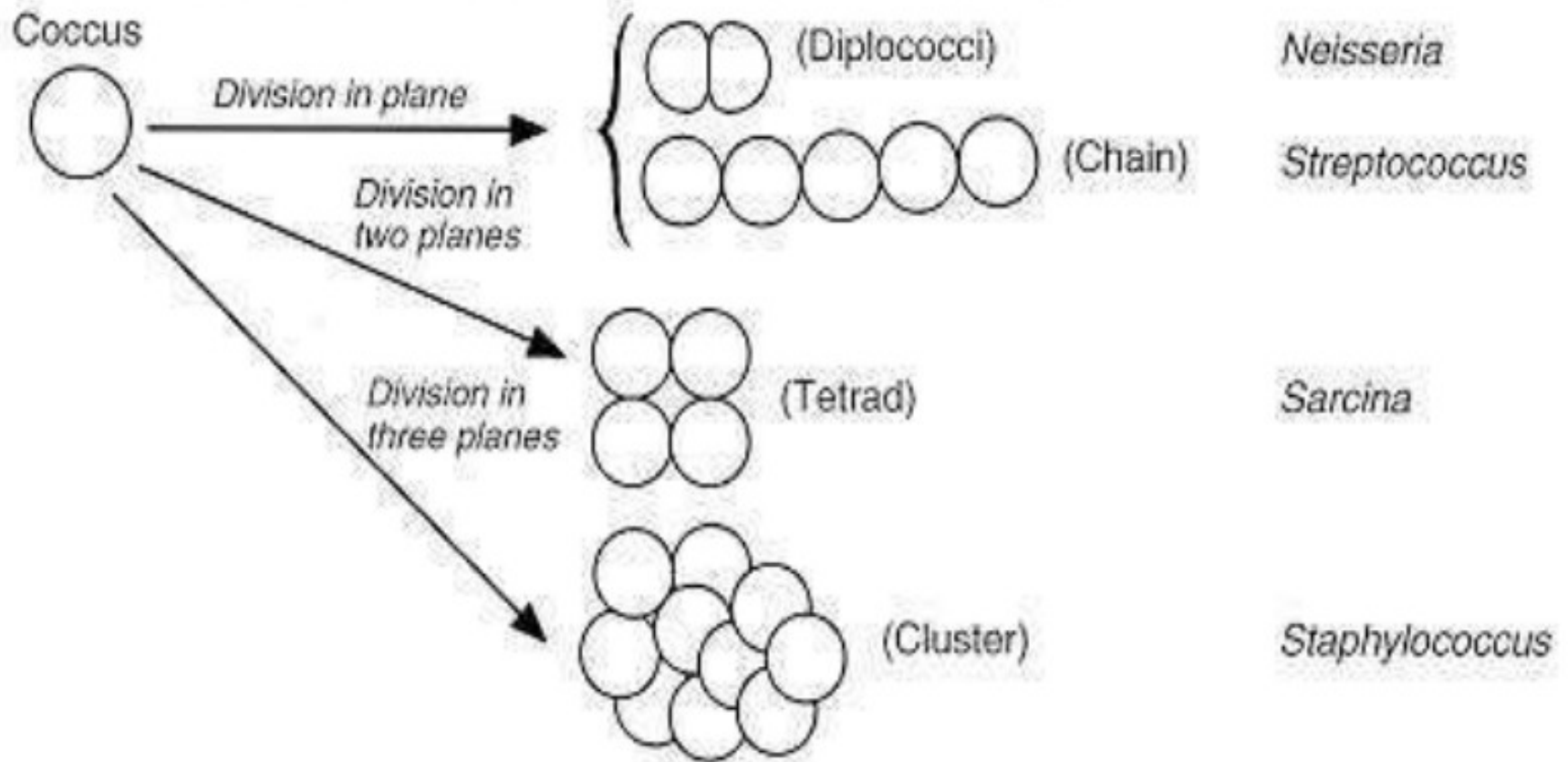
- Cell wall of certain organisms contain long-chain fatty acids (mycolic acids)
- Acid-fast stains are used to stain bacteria that have a high lipid and wax content in their cell walls
- The cell wall is treated to allow penetration of the dye either by **heat** (Ziehl-Neelsen method) or by **a detergent (Kinyoun method)**
- Acid fastness is a property of the mycobacteria
- Microorganisms that retain the acid-fast stain are called *acid-fast*.
- Three dyes are applied
 - Carbol fuchsin a red dye
 - Acidified alcohol
 - methylene blue

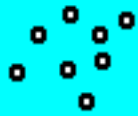


Shape of Bacteria

- Depending on their shape, bacteria are classified into several types
- **Cocci:** oval or spherical cells ,
 - arranged in pairs (e.g., pneumococci, meningococci, and gonococci),
 - tetrads (micrococci),
 - chains (e.g., streptococci), and
 - clusters (e.g., staphylococci).
- **Bacilli:** rod shaped
 - *Coccobacilli*: Length of the bacteria is approximately the same as its width, e.g., *Brucella*.
 - *Streptobacilli*: These are arranged in chains, e.g., *Streptobacillus*.
 - *Comma shaped*: They exhibit curved appearance, e.g., *Vibrio*.
 - *Spirilla*: They exhibit rigid spiral forms, e.g., *Spirillum*.
- **Spirochetes:** slender, flexuous spiral forms, e.g., *Treponema*

Cell arrangements





Coccus



Streptococcus



Staphylococcus



Diplococcus



Tetrad



Sarcina



Bacillus



Diplobacillus



Streptobacillus



Coryneform
Bacillus



Spirillum



Vibrio

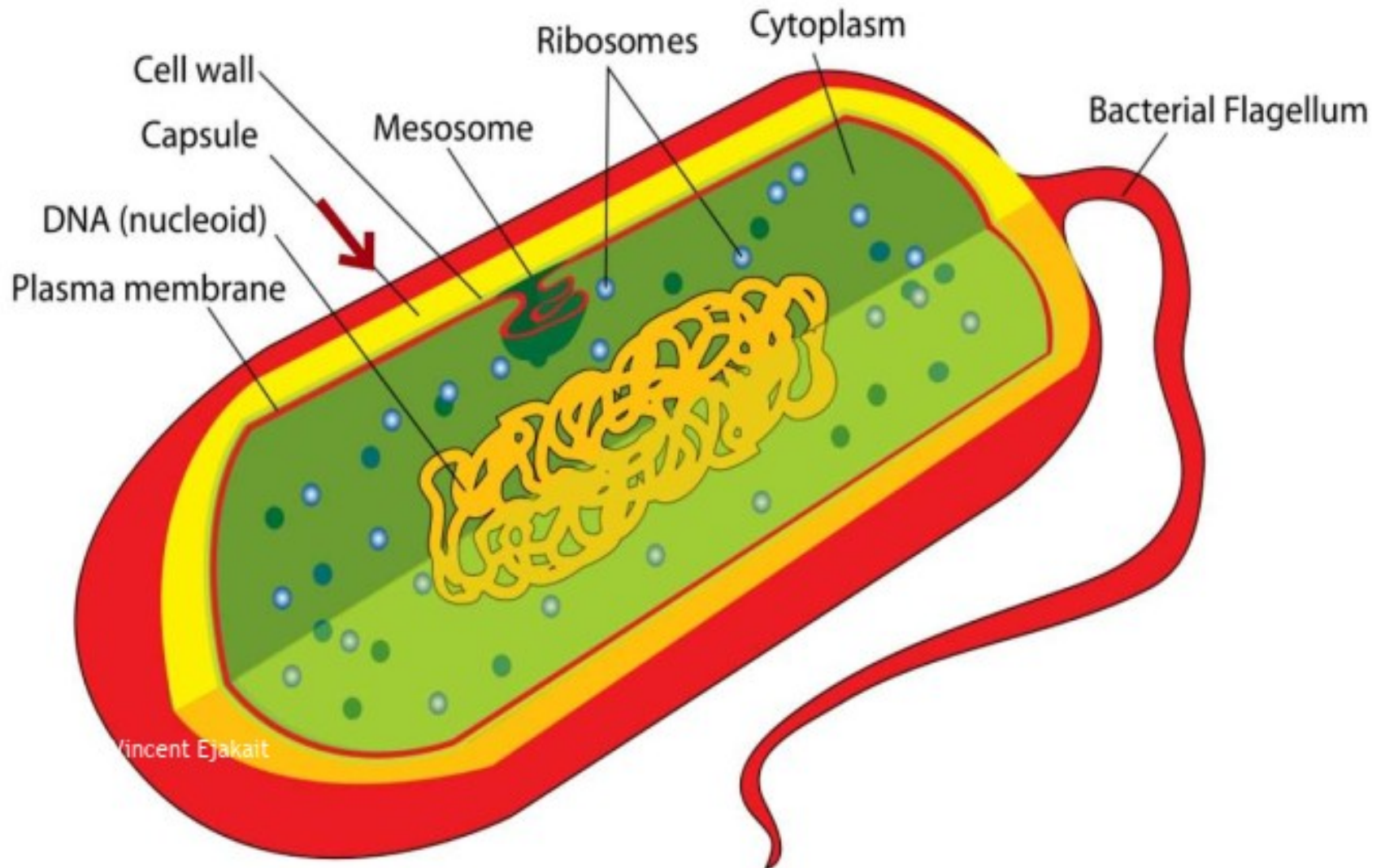


Spirochete

Structural component of Bacteria

- Bacterial structure is categorized to three levels
 - Cell envelope proper: **Cell wall** and **cell membrane**.
 - Cellular element enclosed within the cell envelope
 - Mesosomes, ribosomes, nuclear apparatus, polyamines and cytoplasmic granules
 - Cellular element external to the cell envelope: **Flagellum**, **Pilus** and **Fimbriae**

Bacteria cell structure



Cell envelope

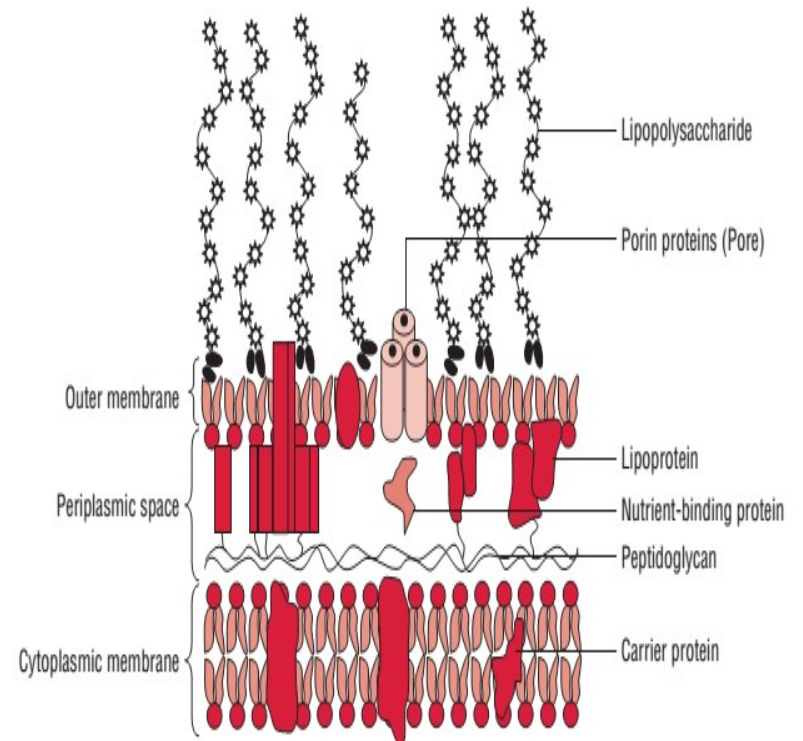
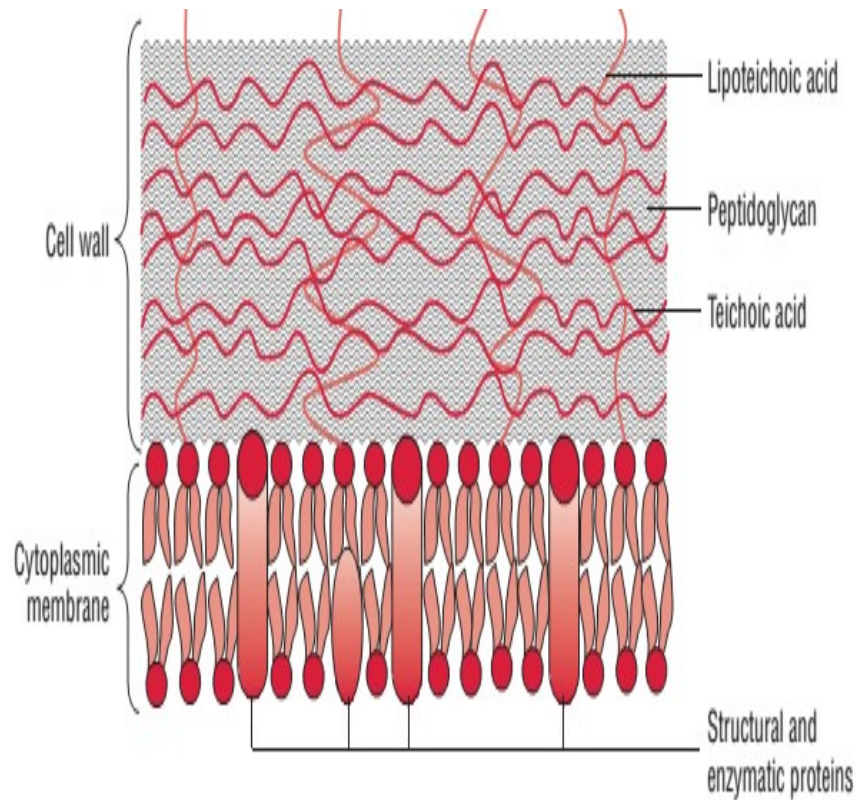
- Is an exceedingly **complex** part of the cell
- Provides a structural and physiological barrier between the protoplasm and the external environment.
- The complexity is appreciated by understanding the three important principles of bacterial functional anatomy
 - ⌚ Responsible for many cellular processes that are the province of the internal organelles
 - ⌚ Primary site of functions that protect the bacterial cell against chemical and biological threats in its environment
 - ⌚ With certain appendages make possible the colonization of surfaces by bacteria

Cell wall

- Rigid wall that determines the shape of the bacterial cell, exceptions.
- Made of a huge molecule called peptidoglycan (murein).
- Peptidoglycan occurs only in the Bacteria,
- PG is made up of two alternate repeating amino sugar and peptides.
 1. The amino sugars are
 - A. N – acetyl glucose amine (NAG)
 - B. N – acetyl muramic acid (NAM)
 2. Peptides – which consist of 4 amino acids
 - L -alanin
 - D-isoglutamic acid
 - L-lysine (or di aminopimelic acid)
 - D- alanin

Cell wall

- In GP the PG forms a thick mesh like layer that retains the blue dye of the **Gram stain** .
- In GN the PG layer is very thin (only one or two molecules deep)
- In gram-positive bacteria the cell wall is composed mainly of
 - a thick **peptidoglycan** mesh work
 - interwoven with other polymers called **teichoic acids**
 - some **proteins** or **lipids**
- Gram-negative bacteria have a complex cell wall that is composed of multiple layers
 - **outer membrane** layer lies on top of a **thin peptidoglycan** layer
 - **Lipopolysaccharides** often called **endotoxins**
 - **Periplasmic** space
 - **Phospholipids**



Comparison of the gram-positive and gram-negative bacterial cell walls

Differences between Gram-positive and Gram-negative bacteria cell wall

Characters	Gram-positive cell wall	Gram-negative cell wall
Thickness	15–80 nm	2 nm
Lipid content	2–5%	15–20%
Teichoic acid	Present	Absent
Variety of amino acid	Few	Several
Aromatic amino acid	Absent	Present
Action as endotoxin	No	Yes
Sulfur-containing amino acid	Absent	Present
Treatment with lysozyme	Protoplast	Spheroplast

Functions of cell wall

-  Provides shape and rigidity to the bacterium
-  Protects from environment
-  Provides staining characteristics to the bacterium
-  Contains receptor sites for phages/complements
-  Site of action of antibody and colicin
-  Contains toxic components to host

Cell wall of acid-fast bacilli

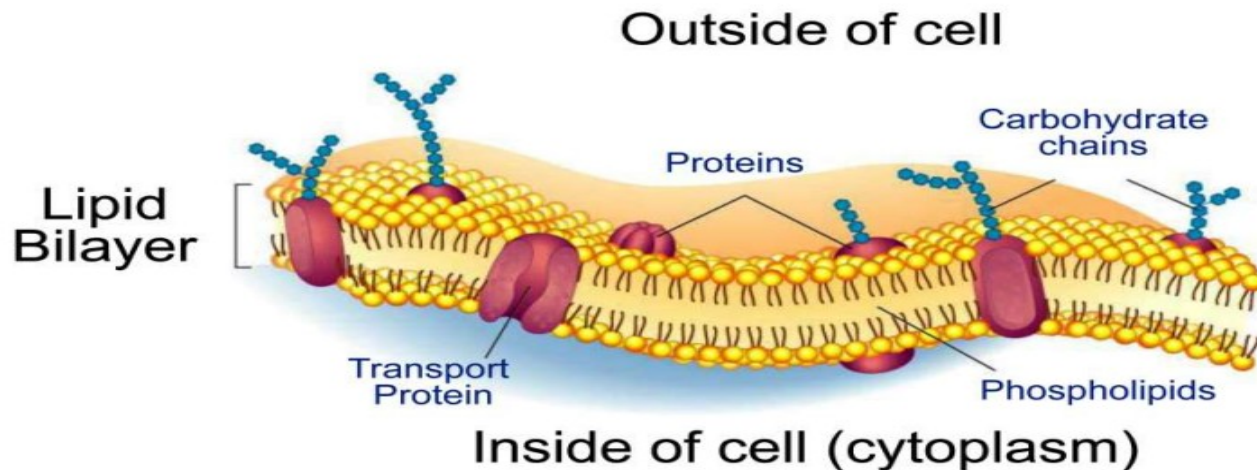
- Contains large amounts of waxes known as **mycolic acids**
 - peptidoglycan and an outer asymmetric lipid bilayer
 - Arabinoglycan
 - Other extractable lipids
- This hydrophobic structure renders these bacteria resistant
 - Harsh chemicals and staining reagents
- During staining, if dye is introduced into these cells by brief heating or treatment with detergents
 - resist decolorization by sulfuric acid or acid alcohol (acid-fast organisms)

Cell membrane

- Also named as **cytoplasmic membrane** or **plasma membrane**
- Which separates the inside of the cell from its external environment
- Acts as a semipermeable membrane regulating the inflow and outflow of metabolites
- Composed of 60% protein, 20-30% lipids and 10-20% carbohydrate
- The cytoplasmic membrane carries out many necessary cellular functions
 - **Energy generation, Synthesis of cell wall components**
 - **Protein secretion**
 - **Chromosome segregation, and Assists DNA replication**
 - **Efficient active transport of nutrients and waste products**

Cell membrane

- Basic structure of cell membrane is **phospholipid–protein** bilayer, usually lacking sterols
- Bacterial cell membrane is exceptionally rich in proteins (up to 70% of its weight)
- Protein-to-lipid ratios of bacterial plasma membranes are approximately 3: 1
- Cytoplasmic membrane acts as a diffusion barrier for water, **ions**, **nutrients**, and transport systems.



Cellular element enclosed within the cell envelope

- ❑ **Mesosomes**, formed by an invagination of the plasma membrane, more prominent in GP than in GN organisms
 - involve in DNA segregation, respiratory enzyme activity, secretion of proteins and active transport
- ❑ **Ribosomes**, a cytoplasmic nucleoprotein particle whose main function is to serve as the site of mRNA translation and protein synthesis
 - RNA(70%) and proteins(30%) and constitutes 90% of the RNA and 40% of the total protein
 - The ribosome monomer is **70s** with two subunits, 30s and 50s
- ❑ **Cytoplasm**, or protoplasm, is a site of cell growth, metabolism, and replication.
 - It is a gel-like matrix composed of water, enzymes, nutrients, wastes, gases, chromosome, and plasmids.
 - provision of nutrient

Nuclear material

- Concentrated in the cytoplasm as a nucleoid
- The nucleoid consists of one long double-stranded circular DNA molecule (chromosome).
- The chromosome serves as the **control center** of the bacterial cell, carries the **genetic information** needed for producing several thousand enzymes and other proteins.
- The size of the chromosome varies according to species.
- The prokaryotic nucleoid is considered **primitive nucleus**,
 - it is not surrounded by a nuclear membrane
 - It does not have a definite shape, and has little or no protein material.
- Apart from nucleus, the bacteria may have some extra chromosomal genetic material in the form of DNA, which is known as **Plasmid**.

Cellular element external to the cell envelope

- **Capsule:** is a polysaccharide layer that completely envelopes the cell
 - used to classify to serotypes based on the fine chemical structure of this polysaccharide
- **Features of capsule**
 - Usually weakly antigenic
 - Not necessary for viability
 - Endows virulence
 - Protects from phagocytosis
 - Capsulated strains are invariably non-motile
 - Visualized by negative staining and capsule staining
 - Detected by quellung phenomenon
 - It can help in attachment to surfaces

Surface Appendages

❑ **Flagella:** are rotating helical protein(flagellin) responsible for locomotion

❑ Functions of Flagella

- Movements
- Sensation
- Signal transduction
- Adhesion
- Flagella are generally accepted as being important **virulence factors**

❑ Flagellar arrangements

- **Atrichous:** Bacteria with no flagellum.
- **Monotrichous:** Bacteria with single polar flagellum. eg. *Vibrio cholera*
- **Lophotrichous:** Bacteria with bunch of flagella at one pole.
- **Amphitrichous:** Bacteria with flagella at both poles.
- **Peritrichous:** Bacteria with flagella all over their surface.

❑ Parts of Flagella

Flagellum

a. Filament or shaft

It is external to the cell surface twisted helically and connected to the hook at the cell surface.

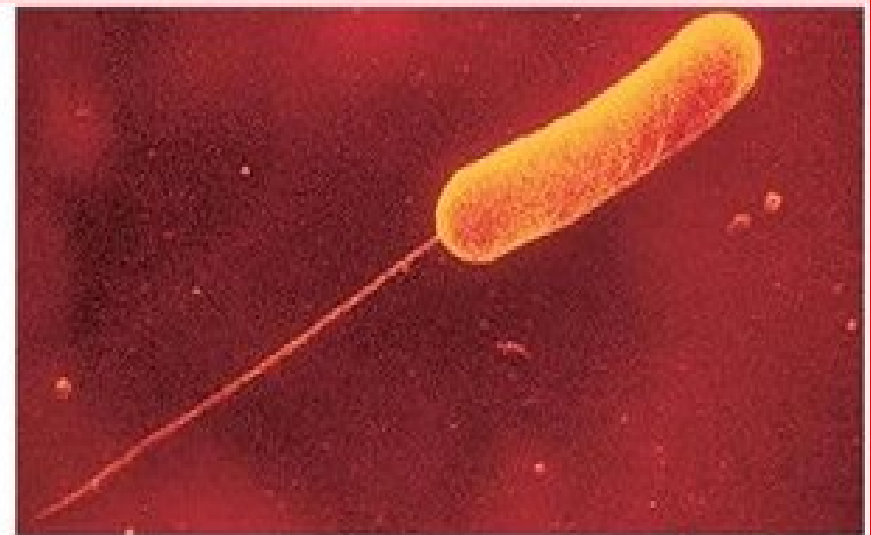
b. Hook – connect the basal body with the filament

c. Basal body – embedded in cell envelope and consists of a small, central rod surrounded by a series of ring

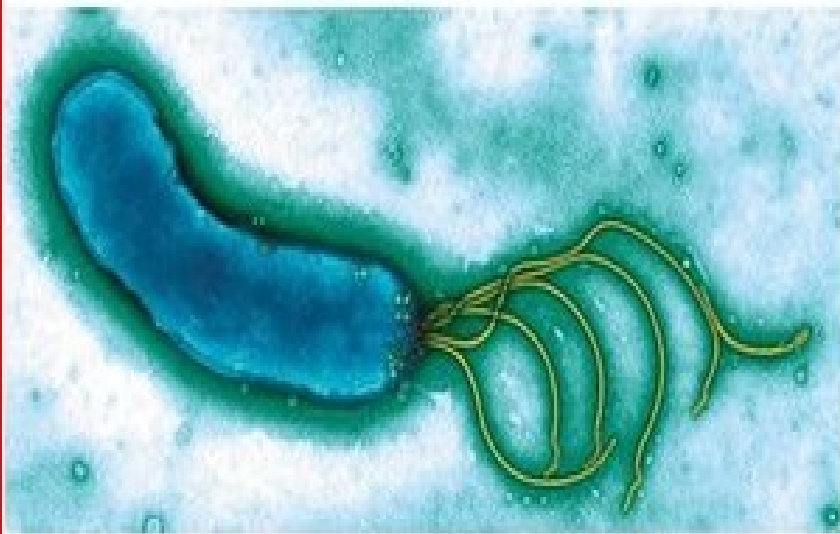
- L-ring – attach to Lipopolysaccharide (LPS) membrane
- P-ring – attach to peptidoglycan
- S-ring – located in the periplasmic space above the plasma membrane
- M-ring – attach to cytoplasmic membrane



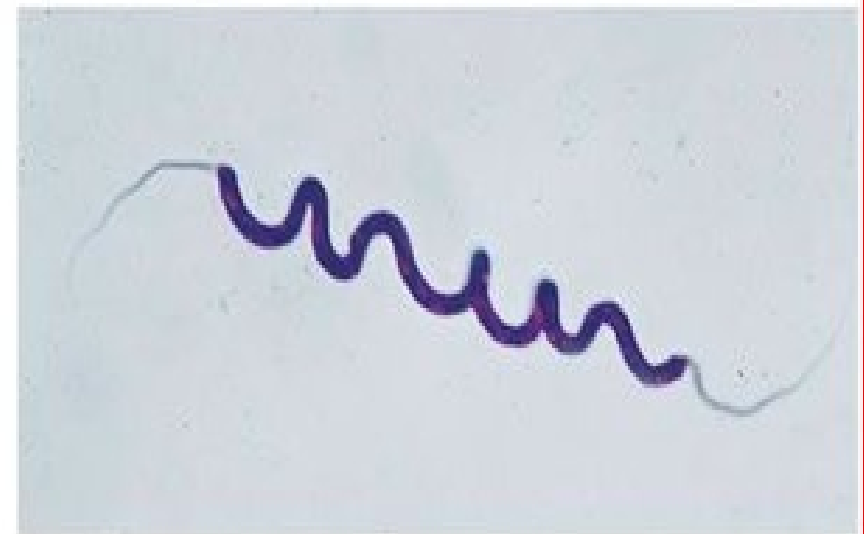
(a) Peritrichous



(b) Monotrichous and polar



(c) Lophotrichous and polar



(d) Amphitrichous and polar



Pilli and Fimbriae

- interchangeable terms used to designate short, hair-like structures (finer filaments) on the surfaces of prokaryotic cells.
- are extruding from the **cytoplasmic membrane**
- are shorter and stiffer than flagella, and slightly smaller in diameter.
- They are involved in the bacterial **conjugation**, **attachment** to the surface and **motility**
- Like flagella, they are composed of protein called **pilin** arranged in helical strand

Two functional types of pili may occur independently or together on some cell.

I. Common pili

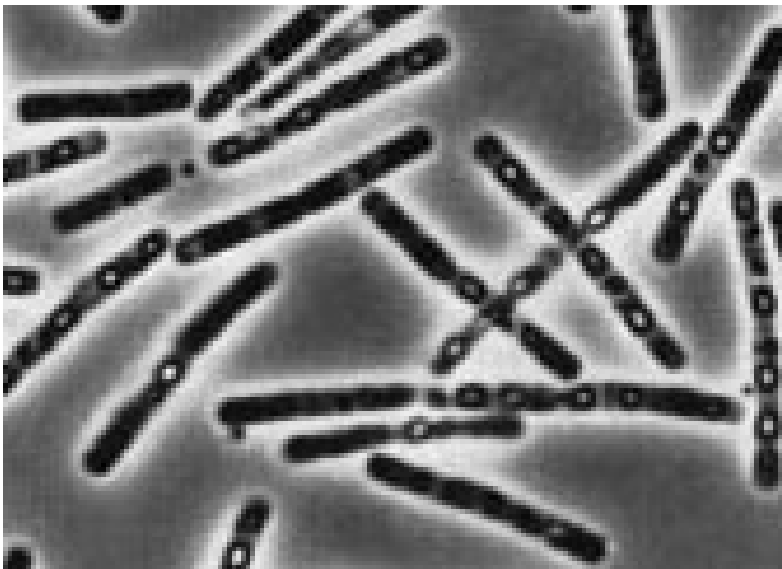
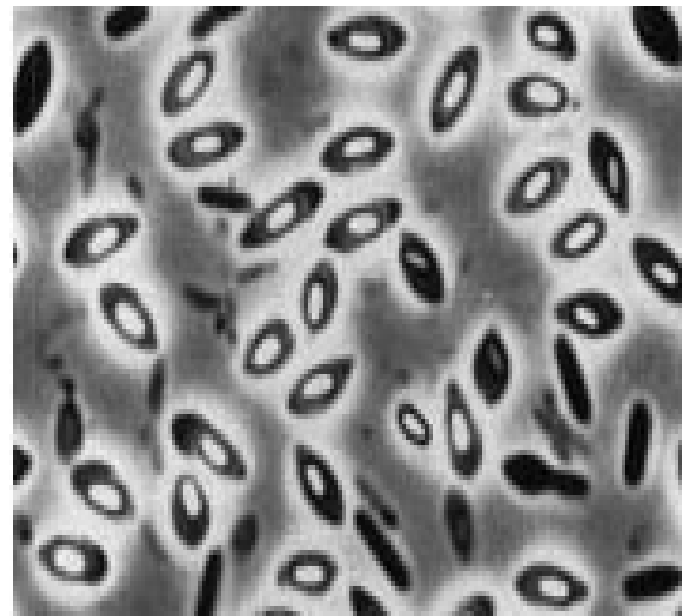
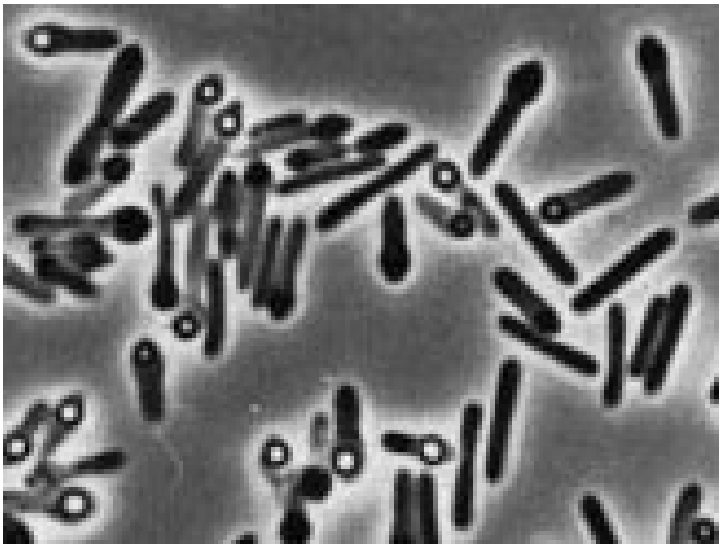
- almost always called **fimbriae**
- Help for attachment of bacteria to epithelial cell
- They considered as **virulence factor** in some species of bacteria, allow pathogens to attach to (colonize) tissues and to resist attack by phagocytic white blood cells.

II. Sex pili or F pili

- occur less commonly
- appear to be specifically involved in bacterial conjugation, i.e transfer of genetic material (DNA) from one bacterium to another.

Bacterial spores

- Under conditions of **limited** supply of **nutrition**, vegetative forms of certain bacteria especially gram-positive bacilli and actinomycetes form highly resistant and dehydrated forms, which are called **endospores**.
- These endospores are capable of survival under **adverse conditions**
 - heat, drying, freezing, radiation, and actions of toxic chemicals.
- Spores are usually smooth walled and ovoid, in some species it is spherical.
- It does not take up ordinary stains
- It looks like areas of high refractivity under light microscope



Different arrangements of spores in the bacteria

Bacterial toxins

- **Toxins** are a very specific virulence factor produced by some bacterial pathogens
- There are two categories of toxins, the **exotoxins** and the **endotoxins**
- In order for a bacterium to be virulent it must have capabilities that allow it to infect a host (**virulence factors**)
 - physical structures
 - chemical substances

Exotoxins

- heat-sensitive soluble proteins that are products released extracellularly as the organism grows
 - can spread throughout the host's body, causing damage distant from the original site of infection
- associated with specific diseases often carried on plasmids or by prophages
- There are three categories of exotoxins:
 - Type I: cell surface-active: bind to cell receptors and stimulate cell responses (**superantigen toxins**)
 - Type II: membrane-damaging :exert their effect on the host cell membrane (**cytolytic toxins**)
 - Type III: intracellular-gain access to a particular host cell and stimulate a reaction within the target cell (**A-B toxins**)

Endotoxins

- Made by gram negative bacteria as a component of the lipid A part of LPS
 - referred as cell-associated toxins
 - Heat-stable and is only released when the bacterial cell is lysed
 - A massive release of endotoxin in a host can cause endotoxin **shock**, which can be deadly
 - act at, or near, the bacterial growth site
- Possesses multiple biological properties
 - induce fever
 - Initiate the complement and blood cascades
 - activate B lymphocytes
 - stimulate production of tumor necrosis factor (TNF)

Classification of bacteria

- Bacterial classification depends on the following characteristics
 - ✓ Morphology and arrangement
 - ✓ Staining
 - ✓ Cultural characteristics
 - ✓ Biochemical reactions
 - ✓ Antigenic structure
 - ✓ Base composition of bacterial DNA

Cont.....

- Morphology and staining of bacteria are the commonly used characteristics to classify bacteria.
- **Morphological features** Including **shape** and **color** of bacterial colonies
 - Colonies; color, size, shape, and smell
- **Biochemical tests** are important in the identification of a genus and species of bacteria
- **Biochemical tests** include
 - nutrients a bacteria can use
 - products of its metabolism
 - Response to specific chemicals
 - presence of particular characteristic enzymes
 - requirement for anaerobic or aerobic environments

TAXONOMIC CLASSIFICATION OF ORGANISMS

- **Taxonomy**; the theory and practice of classifying organisms
- Is the science of classification, identification, and nomenclature.
- For classification purposes, organisms are usually organized into **subspecies**, **species**, **genera**, **families**, and **higher orders**.
- **Classification**: orderly arrangement of bacteria into groups
 - There is nothing inherently scientific about classification
- **Identification**: is the practical use of classification criteria to distinguish certain organisms from others
- **Nomenclature**: is the means by which the characteristics of a species are defined and communicated among microbiologists.

Approaches to Taxonomy

- **Numerical Approach** (computer or phenetic taxonomy)
 - morphologic and biochemical pattern
- **Phylogenetic Approach**
 - evolutionary relatedness among various groups of microorganisms
 - compare each gene sequence in a given strain with the gene sequences for every known species
- **Polyphasic Approach**
 - genotypic, chemotypic and phenotypic features used
 - phenotypic grouping of strains by morphological, biochemical
 - DNA relatedness by phylogenetic analysis
 - reexamination of the biochemical characteristics of the DNA relatedness groups